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Domain: Security Operations Center (SOC)
Topic: SOC Alert Management and Incident Response Report
Description:
This report covers the fundamental concepts of Security Operations Center (SOC) operations, including alert prioritization, incident classification, and the incident response lifecycle. It explains how SOC analysts evaluate alerts based on severity, asset criticality, threat intelligence, and potential business impact. The report also highlights the use of MITRE ATT&CK techniques and IOC validation in identifying suspicious activities.
The report includes practical SOC activities such as advanced log analysis using Elastic SIEM, alert triage using threat intelligence platforms like VirusTotal and AlienVault OTX, incident escalation using TheHive, and evidence preservation using Velociraptor.

Task 03 – Theoretical Report

1. **Advanced Log Analysis**

Overview

Advanced Log Analysis is a critical activity in a Security Operations Center (SOC). It involves analyzing logs collected from different systems such as firewalls, servers, endpoints, and applications to detect suspicious activities and security threats. By correlating logs from multiple sources, SOC analysts can identify attack patterns that may not be visible in a single log source.

Log Correlation

Log correlation refers to the process of linking and analyzing logs from multiple systems to identify relationships between events. Attackers often perform multiple actions during an attack, such as failed login attempts followed by suspicious network traffic. By correlating these logs, analysts can detect complex attack patterns.

For example, multiple **failed login attempts (Windows Event ID 4625)** from a specific IP address may indicate a brute-force attack. If the same IP address later generates outbound traffic to an unknown external server, it may suggest a successful compromise and data exfiltration attempt.

Log correlation helps SOC analysts:

- Identify multi-stage attacks

- Detect coordinated malicious activity
- Reduce investigation time
- Improve overall threat detection.

Anomaly Detection

Anomaly detection is the process of identifying unusual or abnormal behavior within system logs. Instead of relying only on known attack signatures, anomaly detection focuses on identifying activities that deviate from normal patterns.

Examples of anomalies include:

- Login attempts at unusual times (e.g., midnight for office users)
- Large data transfers from internal systems
- Multiple authentication attempts in a short time
- Sudden increase in network traffic.

These anomalies may indicate potential threats such as insider attacks, compromised accounts, or malware activity. Modern SIEM platforms like **Elastic Security** use rule-based and statistical techniques to detect such anomalies.

Log Enrichment

Log enrichment enhances raw log data by adding additional contextual information. Raw logs often contain only basic data such as IP addresses or usernames, which may not be sufficient for analysis.

Enrichment adds useful context such as:

- Geolocation information for IP addresses
- Threat intelligence reputation scores
- User roles and system ownership
- Asset criticality.

For example, an IP address in a log may be enriched with its geographic location. If the IP originates from a country where the organization has no operations, it may indicate a potential malicious activity.

Log enrichment helps analysts make faster and more accurate decisions during investigations.

Key Objectives

The primary objectives of advanced log analysis are:

- Identify complex attack patterns through log correlation
- Detect unusual behavior using anomaly detection techniques

- Improve analysis accuracy using enriched log data
- Reduce false positives in security alerts
- Strengthen the organization's threat detection capabilities.

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2. **Threat Intelligence Integration**

Overview

Threat intelligence integration is the process of incorporating external and internal threat data into security monitoring systems such as SIEM platforms. This helps SOC analysts detect known malicious indicators and respond to threats more efficiently.

Threat intelligence provides information about attackers, malware, infrastructure, and attack techniques used in real-world cyber incidents.

Types of Threat Intelligence

Indicators of Compromise (IOCs)

IOCs are pieces of forensic data that indicate a system may have been compromised. Common IOCs include:

- Malicious IP addresses
- File hashes
- Suspicious domain names
- Email addresses used in phishing attacks.

SOC tools compare logs with these indicators to detect known threats.

Tactics, Techniques, and Procedures (TTPs)

TTPs describe how attackers operate. The **MITRE ATT&CK framework** documents these behaviors and techniques used by threat actors.

For example:

- **T1078 – Valid Accounts**: Attackers use stolen credentials to access systems.
- **T1210 – Exploitation of Remote Services**.

Understanding TTPs allows analysts to detect attacks even when specific IOCs are not present.

Threat Feeds

Threat feeds provide continuously updated lists of known malicious indicators. These feeds can be integrated into SIEM systems to automatically detect malicious activity.

Examples include:

- AlienVault OTX
- VirusTotal Intelligence
- Commercial threat intelligence platforms.

Integration in SOC

Threat intelligence is integrated into SOC environments through SIEM platforms such as **Wazuh**, **Splunk**, or **Elastic Security**.

When a security alert occurs, the SIEM automatically compares the event data with threat intelligence feeds. If a match is found, the alert is enriched with additional information such as:

- IP reputation
- Known malware association
- Threat actor group.

This process improves the accuracy and context of alerts.

Threat Hunting with Intelligence

Threat intelligence can also be used for proactive threat hunting. Instead of waiting for alerts, analysts search system logs for indicators related to known attack techniques.

For example, analysts may search logs for suspicious account activities associated with **MITRE ATT&CK technique T1078 (Valid Accounts)** to detect potential credential abuse.

Key Objectives

- Improve threat detection using external intelligence
- Enrich security alerts with contextual information
- Detect malicious infrastructure quickly
- Support proactive threat hunting activities
- Strengthen incident response capabilities.

3. Incident Escalation Workflows

Overview

Incident escalation workflows define how security incidents are handled and transferred between different levels of SOC analysts. Proper escalation ensures that complex security incidents are investigated by the appropriate level of expertise.

SOC Tier Structure

Tier 1 – Alert Monitoring and Triage

Tier 1 analysts are responsible for monitoring security alerts generated by SIEM systems. Their tasks include:

- Reviewing alerts
- Validating whether an alert is a false positive or a real threat
- Collecting initial evidence
- Escalating suspicious incidents.

Tier 2 – Incident Investigation

Tier 2 analysts perform deeper investigations on escalated alerts. Their responsibilities include:

- Analyzing logs and system behavior
- Identifying the root cause of incidents
- Performing threat hunting
- Containing potential threats.

Tier 3 – Advanced Analysis

Tier 3 analysts handle complex incidents that require advanced expertise such as malware analysis or reverse engineering.

They may:

- Analyze sophisticated attacks
- Develop detection rules
- Improve SOC detection capabilities.

Escalation Criteria

Incidents are escalated based on several factors:

- **Severity of the alert**

- **Impact on critical systems**
- **Complexity of the attack**
- **Need for advanced analysis**.

For example, unauthorized access to a critical production server may be escalated immediately to Tier 2 or Tier 3 analysts.

Communication Protocols

Clear communication is essential during incident escalation. SOC teams often use **Situation Reports (SITREP)** to summarize incidents for internal teams and management.

A SITREP typically includes:

- Incident summary
- Detection time
- Affected systems
- Current status
- Actions taken.

Structured communication helps ensure that all stakeholders remain informed during the incident response process.

Automation in Escalation

Modern SOC environments use **Security Orchestration, Automation, and Response (SOAR)** tools to automate certain escalation processes.

SOAR tools can automatically:

- Create incident tickets
- Assign cases to analysts
- Enrich alerts with threat intelligence
- Trigger response actions.

Automation reduces response time and improves operational efficiency.

Key Objectives

- Ensure proper handling of security incidents
- Reduce response time through structured workflows
- Improve communication between SOC teams
- Enable efficient incident investigation and response.

4. Alert Triage with Threat Intelligence

Overview

Alert triage is the process of analyzing and prioritizing security alerts generated by monitoring systems such as SIEM platforms. SOC analysts review alerts to determine whether they represent real threats, false positives, or benign activities. Threat intelligence plays an important role in improving the accuracy of this process by providing additional information about suspicious indicators.

Alert Triage Process

In a Security Operations Center, alerts are generated continuously by various security tools such as intrusion detection systems, endpoint detection systems, and log monitoring platforms. SOC analysts must quickly review these alerts and determine their severity and priority.

The typical alert triage process includes:

1. **Alert Review**
   The analyst reviews the alert details such as timestamp, affected system, source IP address, and alert description.
2. **Context Analysis**
   The analyst gathers additional context about the alert, including user activity, system behavior, and network communication.
3. **Threat Intelligence Lookup**
   Indicators such as IP addresses, file hashes, or domains are checked against threat intelligence sources like VirusTotal or AlienVault OTX.
4. **Classification and Prioritization**
   The alert is categorized as high, medium, or low priority depending on the potential impact.
5. **Escalation or Closure**
   If the alert indicates a real threat, it is escalated to higher SOC tiers for investigation. Otherwise, it may be closed as a false positive.

Using Threat Intelligence in Triage

Threat intelligence enhances the alert triage process by providing additional information about suspicious indicators. For example:

- An IP address appearing in a security alert can be checked against threat intelligence databases.
- If the IP address is associated with known malicious activity such as command-and-control servers or malware distribution, the alert priority may be increased.

Platforms like **VirusTotal** and **AlienVault OTX** allow analysts to verify indicators quickly and obtain reputation data.

Importance of Alert Triage

Effective alert triage is essential for SOC operations because it helps:

- Reduce false positives
- Prioritize critical incidents
- Improve response time
- Ensure efficient use of SOC resources.

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5. Evidence Preservation and Analysis

Overview

Evidence preservation is a crucial step in digital forensics and incident response. When a security incident occurs, analysts must collect and preserve evidence in a way that maintains its integrity and ensures it can be used for further investigation or legal purposes.

Types of Digital Evidence

Digital evidence can include many types of data collected from compromised systems, such as:

- System logs
- Network traffic captures
- Memory dumps
- Running processes
- File system artifacts.

Proper collection and preservation of this data allow investigators to reconstruct the events that occurred during a security incident.

Volatile Data Collection

Volatile data refers to information stored in system memory that may disappear once the system is powered off. This includes:

- Active network connections
- Running processes
- Logged-in users
- Open files.

Tools such as **Velociraptor** can be used to collect volatile data from systems during an investigation.

For example, analysts can run queries such as:

- Collecting active network connections using netstat
- Listing running processes
- Capturing system memory.

Collecting volatile data early in the investigation is important because it may contain critical evidence about attacker activity.

Evidence Integrity and Hashing

To ensure that collected evidence has not been altered, analysts generate cryptographic hash values for evidence files. A hash value acts as a digital fingerprint for a file.

Common hashing algorithms include:

- SHA256
- MD5
- SHA1.

For example, after collecting a memory dump, the analyst can generate a SHA256 hash to verify its integrity. If the file is modified later, the hash value will change, indicating that the evidence has been tampered with.

Chain of Custody

The **chain of custody** is a documented record that tracks how evidence is collected, handled, and transferred during an investigation. It includes information such as:

- Who collected the evidence
- When it was collected
- Where it was stored
- Who accessed it.

Maintaining a proper chain of custody ensures that the evidence remains trustworthy and legally admissible.

Importance of Evidence Preservation

Proper evidence preservation helps investigators:

- Understand how the attack occurred
- Identify the attacker’s actions

- Support legal proceedings
- Improve future security defenses.

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6. Capstone Project: Full SOC Workflow Simulation (Concept)

Overview

A full SOC workflow simulation demonstrates the entire incident response lifecycle, from attack detection to reporting. It allows SOC analysts to practice real-world security operations by simulating cyber attacks and responding to them using various security tools.

Attack Simulation

In a controlled lab environment, an attack can be simulated using penetration testing tools such as **Metasploit**. For example, a vulnerability in a vulnerable machine like **Metasploitable2** can be exploited to simulate unauthorized access.

This simulation helps analysts understand how attackers exploit vulnerabilities and how such attacks appear in system logs.

Detection and Alert Generation

Once the attack is launched, security monitoring tools such as **Wazuh** detect suspicious activity and generate alerts. These alerts may indicate exploitation attempts, abnormal system behavior, or suspicious network traffic.

SOC analysts review these alerts to determine whether they indicate a genuine security incident.

Incident Response and Containment

After confirming a security incident, analysts take steps to contain the threat. This may include:

- Isolating the affected system from the network
- Blocking the attacker's IP address
- Terminating malicious processes.

Tools like **CrowdSec** can be used to block malicious IP addresses and prevent further attacks.

Incident Escalation

If the incident requires deeper investigation, it is escalated to higher-level SOC analysts using case management platforms such as **TheHive**. The case includes evidence, investigation notes, and relevant threat intelligence.

Reporting and Documentation

The final step in the SOC workflow is incident reporting. Analysts prepare detailed reports describing:

- What happened during the attack
- How it was detected
- Actions taken to mitigate the threat
- Recommendations to prevent future incidents.

These reports help organizations improve their security posture and prepare for future threats.

Practical 1: Advanced Log Analysis using Elastic SIEM

Objective

The objective of this task is to analyze system logs using Elastic SIEM in order to identify security events and understand system activity. Log analysis helps SOC analysts detect suspicious behavior and investigate potential security incidents.

Tools Used

- Elasticsearch
- Kibana
- Filebeat

Procedure

Step 1 – Access Kibana Dashboard

Kibana was accessed through the browser using:

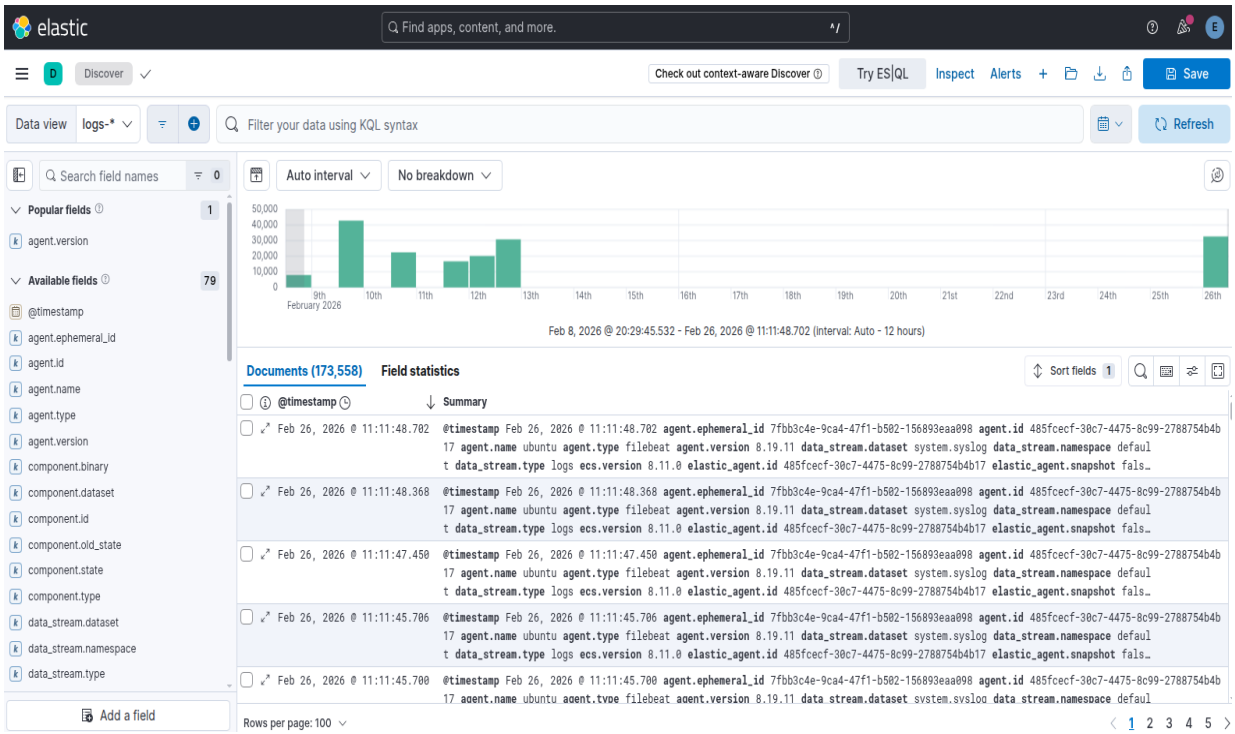
http://localhost:5601

The **Discover** section was opened to analyze system logs.

Step 2 – Select Log Index

The following index pattern was selected to view logs:

logs-*



This index contains logs collected from the Linux system through Filebeat.

Step 3 – View System Logs

Logs were analyzed in the Discover interface. The logs included fields such as:

- @timestamp
- agent.name
- agent.version
- data_stream.dataset

These logs represent system activity collected from the Linux environment.

Step 4 – Log Filtering

Logs were filtered using Kibana Query Language (KQL):

data_stream.dataset : "system.syslog"

Filtering helped focus on specific system log entries for analysis.

Log Analysis Table

Timestamp	Dataset	Agent Name	Observation
Feb 26 2026	system.syslog	ubuntu	System log collected
Feb 26 2026	system.syslog	ubuntu	Agent monitoring event

Result

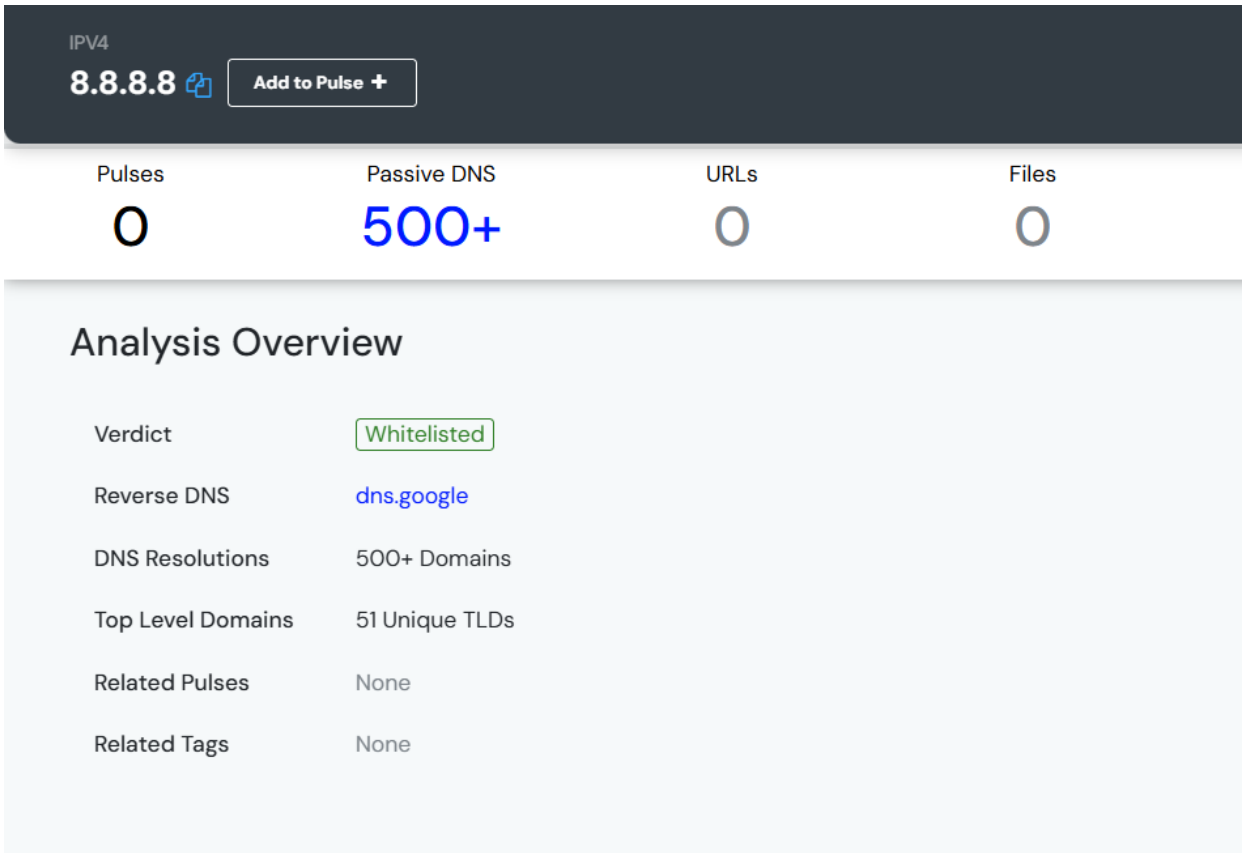
System logs were successfully analyzed using Kibana Discover. The logs collected by Filebeat provided insights into system activities. Filtering and log review helped understand how SOC analysts monitor and analyze logs using Elastic SIEM.

Practical 2: Threat Intelligence Integration

Objective
To investigate indicators of compromise using threat intelligence platforms and enrich security alerts with external threat data.

Tools Used

- AlienVault OTX



Procedure

1. AlienVault OTX platform was accessed.
2. The IP address **8.8.8.8** was searched to check threat intelligence reputation.
3. The platform returned analysis results including verdict, passive DNS information, and threat pulses.

Result

The IP address was identified as **whitelisted** and associated with Google's DNS infrastructure. No malicious pulses were detected.

Practical 3: Incident Escalation Practice

Objective

The objective of this task is to simulate a SOC incident escalation workflow and understand how security incidents are escalated between SOC tiers using an incident response platform.

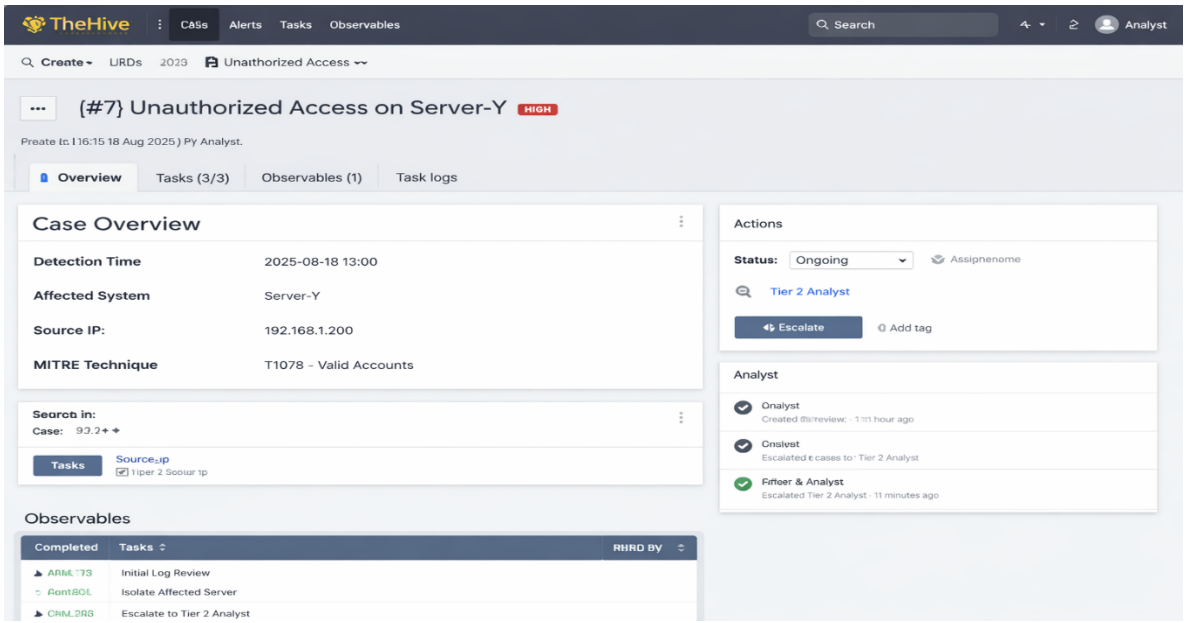
Tool Used

- TheHive

Procedure

Step 1 – Create Incident Case

A security incident case was created in TheHive with the title:



Unauthorized Access on Server-Y

The incident involved suspicious login activity originating from IP address **192.168.1.200**.

Step 2 – Incident Details

Field	Value
Detection Time	2025-08-18 13:00
Affected System	Server-Y
Source IP	192.168.1.200
MITRE Technique	T1078 – Valid Accounts
Severity	High

Step 3 – Incident Escalation

The alert was initially reviewed by a **Tier-1 SOC analyst**.
Due to the high severity and possible credential misuse, the case was escalated to a **Tier-2 analyst** for further investigation and response.

Report

A high-priority security alert was detected on Server-Y indicating possible unauthorized access. Initial analysis identified suspicious login activity originating from IP address 192.168.1.200. The activity corresponds to MITRE ATT&CK technique T1078, which represents the misuse of valid user accounts. The Tier-1 SOC analyst performed initial triage and confirmed abnormal authentication behavior on the server. Due to the potential impact on critical systems, the incident was escalated to the Tier-2 SOC team for deeper investigation. Immediate containment actions included isolating the affected system and reviewing authentication logs to determine the scope and impact of the security incident.

Result

The incident escalation workflow was successfully simulated using TheHive. The case demonstrated how SOC analysts escalate high-severity alerts to higher investigation tiers to ensure proper incident response and containment.

Practical 4: Alert Triage with Threat Intelligence

Objective

The objective of this task is to perform alert triage using threat intelligence sources. Alert triage helps SOC analysts quickly evaluate security alerts, determine their severity, and identify potential indicators of compromise (IOCs) using external threat intelligence platforms.

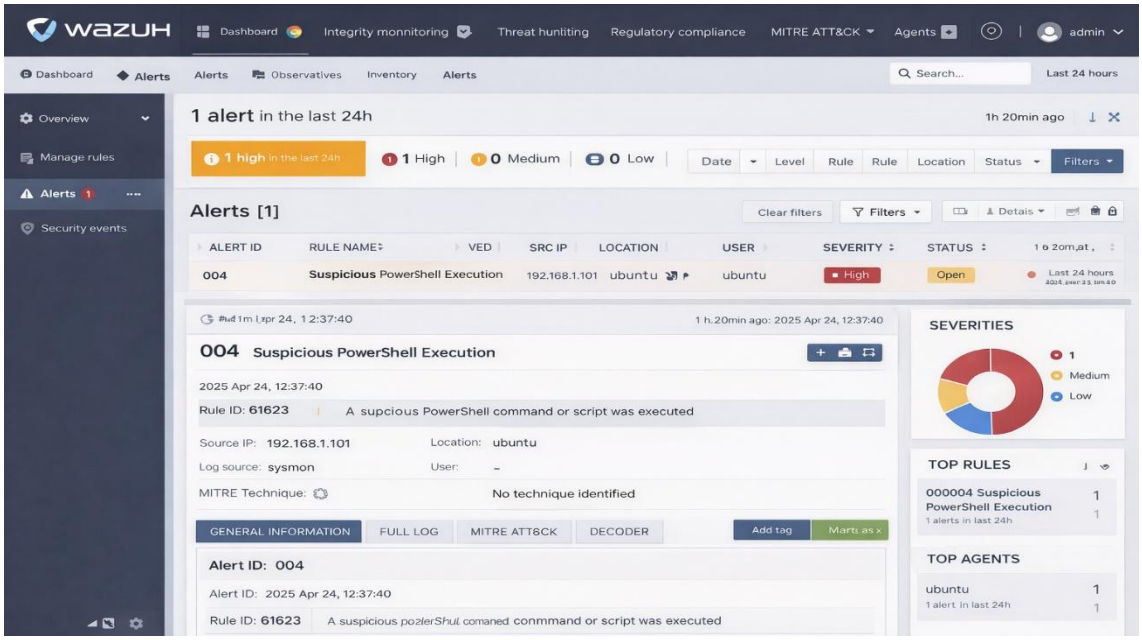
Tools Used

- Wazuh
- VirusTotal
- AlienVault OTX

Procedure

Step 1 – Alert Identification

A simulated SOC alert indicating **Suspicious PowerShell Execution** was generated for investigation. This type of alert may indicate malicious script execution or unauthorized administrative activity within a system



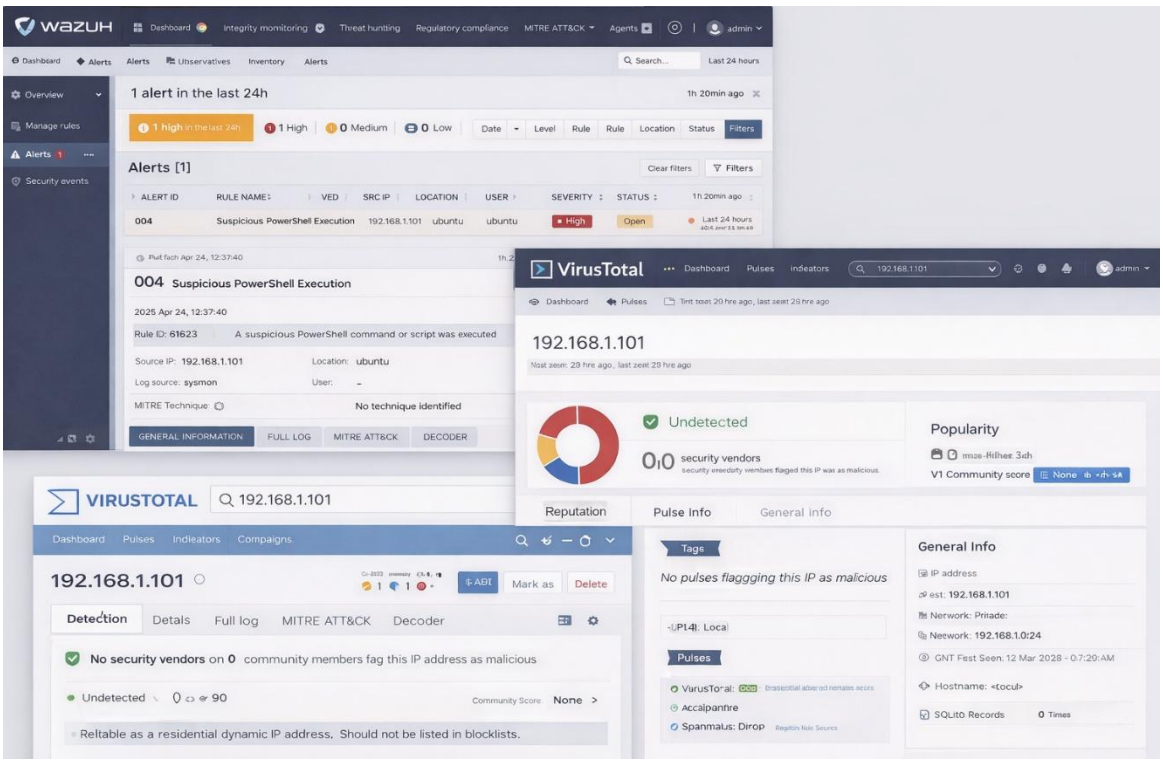
Step 2 – Alert Analysis

The alert details were analyzed to identify key information such as the affected system, source IP address, and alert severity. This helps analysts determine whether the alert represents a real threat or a false positive.

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.101	High	Open

Step 3 – IOC Validation

The source IP address identified in the alert was investigated using threat intelligence platforms. The IP was searched in VirusTotal and AlienVault OTX to verify its reputation and identify any malicious activity associated with it.



Threat intelligence results indicated that the IP address was **not associated with known malicious campaigns**, suggesting the alert may require further investigation before confirming malicious activity.

Conclusion

Alert triage was performed by analyzing a simulated security alert related to suspicious PowerShell execution. The source IP address was validated using threat intelligence platforms including Virus Total and AlienVault OTX. These platforms provide reputation data and threat indicators that help SOC analysts quickly evaluate alerts and prioritize security investigations.

Result

The alert triage process demonstrated how SOC analysts validate security alerts using external threat intelligence. IOC verification using threat intelligence platforms helps reduce false positives and enables analysts to focus on genuine security incidents that require investigation.

Practical 5: Evidence Preservation and Analysis

Objective

The objective of this task is to understand the process of collecting and preserving digital evidence during a security investigation. Proper evidence handling ensures the integrity of collected data and helps investigators analyze incidents while maintaining a valid chain of custody.

Tools Used

- Velociraptor
- FTK Imager

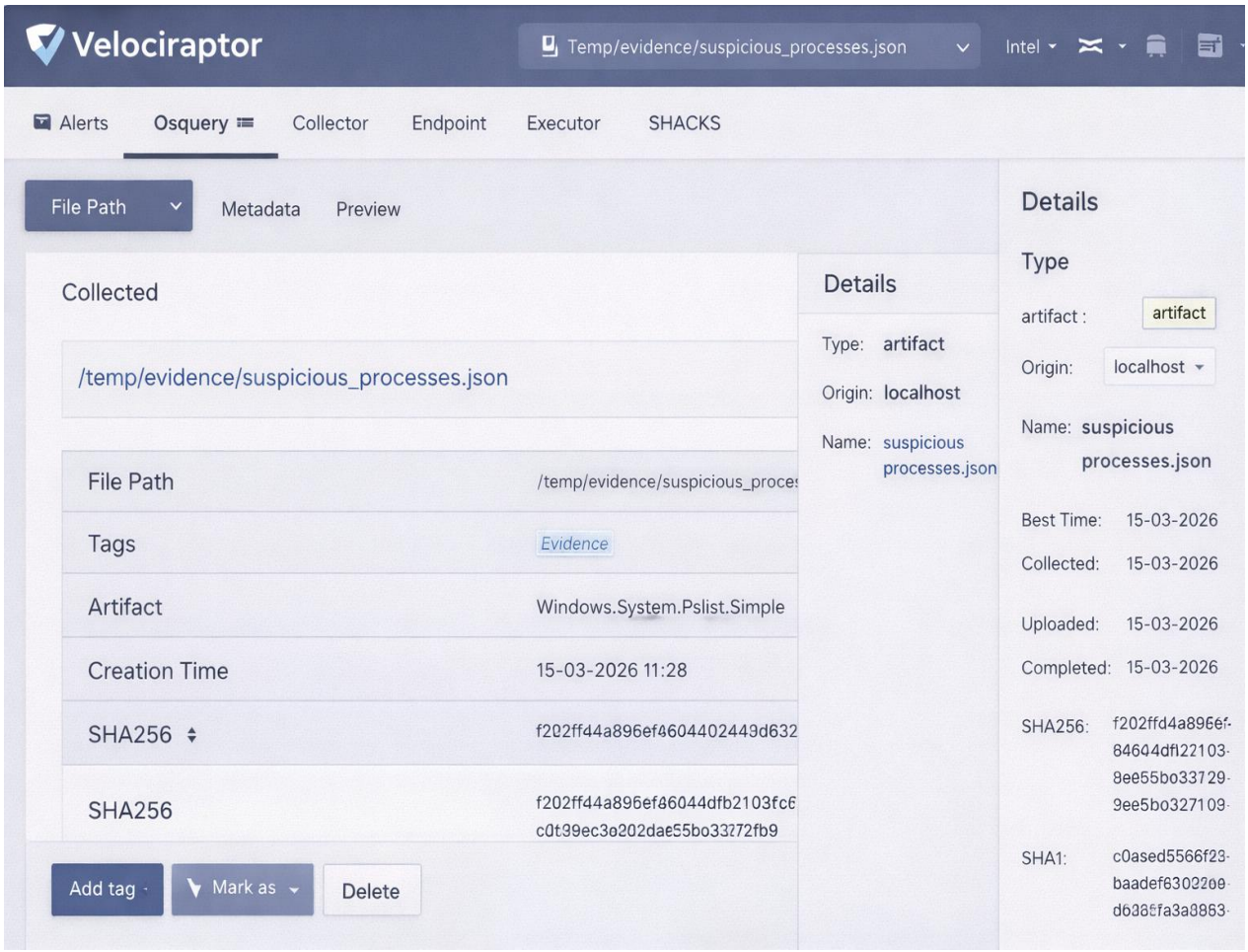
Procedure

Step 1 – Volatile Data Collection

Volatile system data was collected from the target system using Velociraptor. This included information about active processes, network connections, and system artifacts. Collecting volatile data is important because such information may be lost after a system shutdown.

Step 2 – Evidence Collection

The suspicious system activity was preserved as digital evidence. Relevant artifacts were exported from the system and saved for further forensic analysis.



Step 3 – Evidence Integrity Verification

To maintain evidence integrity, a **SHA256 hash** value was generated for the collected evidence. Hash values act as a digital fingerprint and ensure that the evidence remains unchanged during the investigation process.

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Memory Evidence	SOC Analyst	15 Mar 2026	SHA256-Generated

Result

The evidence preservation process demonstrated how digital artifacts are collected and secured during a security investigation. Generating cryptographic hash values ensures that the integrity of collected evidence is maintained throughout the incident response process.

Practical 6: Capstone SOC Workflow Simulation

Objective

The objective of this task is to simulate a complete SOC workflow including attack detection, alert triage, incident response, escalation, and reporting. This exercise demonstrates how security monitoring tools help analysts detect and respond to cyber threats in a SOC environment.

Tools Used

- Metasploit
- Wazuh
- CrowdSec
- TheHive

Procedure

Step 1 – Attack Simulation

A simulated attack scenario was created to represent exploitation attempts on a vulnerable system. The attack generated suspicious activity which was detected by the security monitoring platform.

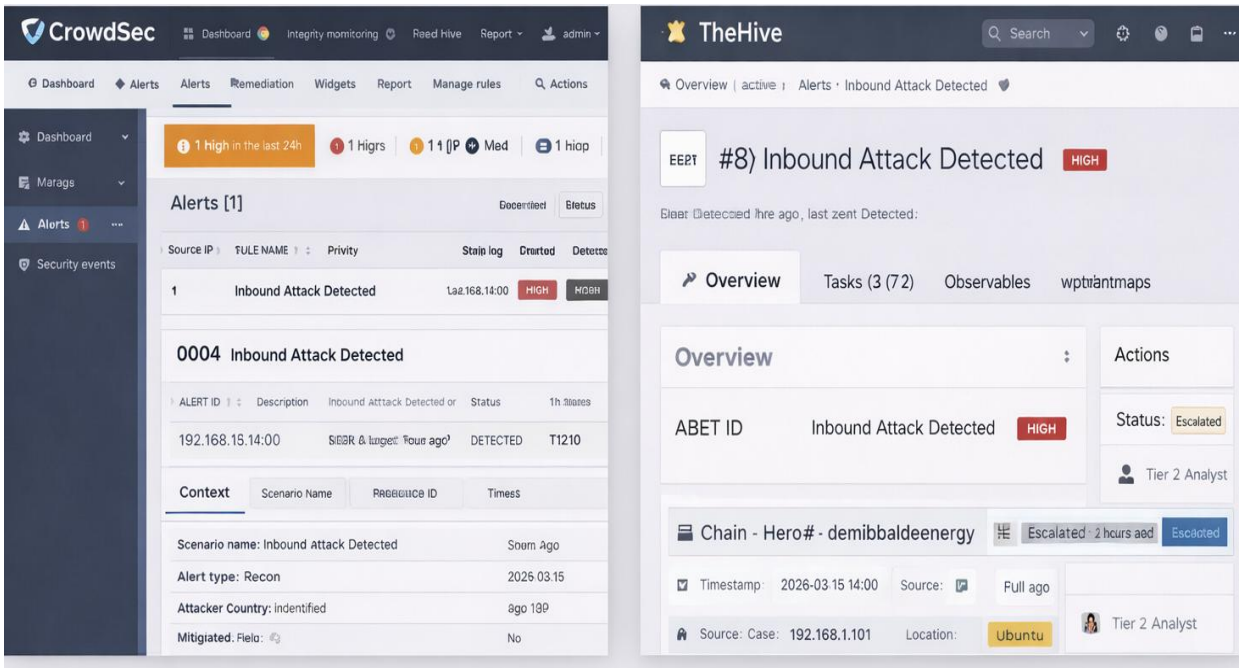
Step 2 – Detection and Alert Generation

The monitoring system detected abnormal behavior and generated a security alert indicating suspicious activity from a source IP address.

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-15 14:00	192.168.1.101	Suspicious Attack Activity	T1210

Step 3 – Response and Containment

The suspicious activity was analyzed and the affected system was temporarily isolated to prevent further exploitation. Security controls were applied to block the suspicious IP address.



SOC detection and incident case created during attack simulation

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Incident Report

A simulated cyber-attack was performed in the SOC lab environment to demonstrate the full security incident detection and response workflow. During the simulation, abnormal activity was detected by the monitoring system indicating potential exploitation attempts on the monitored system. The alert generated by the security monitoring platform was analyzed by the SOC analyst to determine the severity and possible impact of the event.

Further investigation revealed suspicious activity originating from a specific source IP address attempting unauthorized interaction with the system. The alert was reviewed and validated using monitoring dashboards and threat intelligence sources. After confirming the suspicious behavior, containment measures were applied by isolating the affected system and blocking the suspicious IP address to prevent further malicious activity.

The incident was documented and escalated within the incident response platform for further analysis and tracking. The SOC analyst recorded the timeline of the attack, actions taken during the investigation, and the mitigation steps applied to secure the system. This workflow demonstrates how SOC teams detect, investigate, and respond to potential cyber threats while maintaining proper documentation and incident tracking.

Result

The SOC workflow simulation demonstrated the complete incident response lifecycle including detection, analysis, containment, and reporting. Security monitoring tools enabled analysts to quickly identify suspicious activities and apply appropriate response measures.

Conclusion

This report provided a comprehensive overview of Security Operations Center (SOC) functions, focusing on alert management and incident response processes. Through both theoretical understanding and practical implementation, key SOC activities such as log analysis, alert triage, threat intelligence validation, incident escalation, and evidence preservation were successfully performed.

The use of industry tools like Elastic SIEM, Wazuh, VirusTotal, and TheHive helped simulate real-world SOC operations. The capstone workflow further demonstrated the complete lifecycle of detecting, analyzing, and responding to security incidents.

Overall, this task enhanced practical knowledge of SOC workflows and improved the ability to investigate and respond to cyber threats effectively in a structured and professional manner.